

Hierarchical Approximate Min-Sum and Multi-Frame Parallel QC-LDPC Decoder for FPGA Optical Links

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Abstract—In this letter, we present an efficient quasi-cyclic low-density parity-check (QC-LDPC) decoder designed for resource-constrained satellite laser communication terminals. We propose a novel approximate min-sum algorithm that incorporates node grouping and hierarchical comparison, reducing comparator and multiplexer (MUX) usage by 39% and 52%, respectively, with only a negligible 0.02 dB performance degradation. To maximize hardware efficiency, we propose a multi-frame parallel architecture with asynchronous scheduling, achieving near-optimal 99.8% on-chip block RAM (BRAM) utilization and eliminating pipeline stalls. Experimental results demonstrate that the proposed decoder supports 2 Gbps binary phase-shift keying (BPSK) transmission with a minimum required received optical power (ROP) of −53 dBm.

Index Terms—QC-LDPC decoder, approximate min-sum algorithm, FPGA, memory-efficient architecture.

I. INTRODUCTION

WITH the reduced cost of satellite manufacturing and launch, free-space optical (FSO) links have emerged as a promising solution for high-capacity satellite communication systems [1], [2]. However, channel impairments such as vibration, thermal drift, jitter, and noise may severely degrade the link reliability [3]. To ensure reliable high-speed transmission over FSO links, forward error correction (FEC) plays a critical role [4].

LDPC codes are ubiquitous in communication standards such as DVB-S2 and CCSDS [5], [6]. While QC-LDPC codes [7] and Min-Sum algorithms [8], [9] alleviate computational complexity, achieving high-throughput decoding for satellite terminals is still impeded by critical hardware bottlenecks in existing designs. Specifically, representative check node implementations [10] rely on complex sorting networks to

find minimum values, consuming excessive comparators and multiplexers. Furthermore, recent parallel architectures [11], [12] often suffer from pipeline stalls and suboptimal BRAM utilization caused by rigid alignment constraints and fixed block sizes. However, a unified architecture resolving both logic complexity and memory bottlenecks remains lacking.

In this letter, we propose an approximate min-sum decoding algorithm employing node grouping and hierarchical comparison to reduce comparators and shorten the critical path, thereby improving the clock frequency and efficiency. Based on this algorithm, we design a decoder architecture that integrates the intra-frame pipelining and inter-frame resource reuse, supporting multi-frame parallelism with asynchronous scheduling while avoiding conflicts between the check node units (CNU) and variable node units (VNU). A frame-parallel compact-packing memory strategy achieves up to 99.8% BRAM utilization on FPGA. At 2 Gbps BPSK, the decoder operates reliably with a minimum required ROP of −53 dBm and achieves comparator and multiplexer savings of up to 39% and 52%, with less than 0.02 dB performance loss.

II. PRINCIPLES

A. LDPC Decoder Scheme

The NMSA provides a low-complexity approximation of the belief propagation (BP) [13] algorithm by performing computations in the logarithmic domain. Multiplicative operations are replaced by additions, greatly reducing complexity while maintaining near-optimal decoding performance. The decoding starts by initializing the log-likelihood ratio (LLR) of each variable node v_i from the received channel value y_i :

$$L_{vi}^{(0)} = \ln \left(\frac{P(y_n | x_n = 0)}{P(y_n | x_n = 1)} \right) = \frac{2y_i}{\sigma^2} \quad (1)$$

where σ^2 denotes the noise variance of the additive white Gaussian noise (AWGN) channel. This initial message is propagated as $L_{vi \rightarrow cj}^{(0)} = L_{vi}^{(0)}$ and $M_{cj \rightarrow vi}^{(0)} = 0$. At each iteration t , each check node c_j updates its outgoing message to the variable node v_i using the normalized min-sum rule:

$$M_{cj \rightarrow vi}^{(t)} = \alpha \cdot \prod_{vb \in N(c_j) \setminus v_i} \text{sign}(L_{vb \rightarrow cj}^{(t-1)}) \cdot \left| L_{vb \rightarrow cj}^{(t-1)} \right| \quad (2)$$

where $\alpha \in (0, 1]$ is a normalization factor used to mitigate the overestimation inherent in the min-sum approximation. The set $N(c_j) \setminus v_i$ denotes all variable nodes connected to the check

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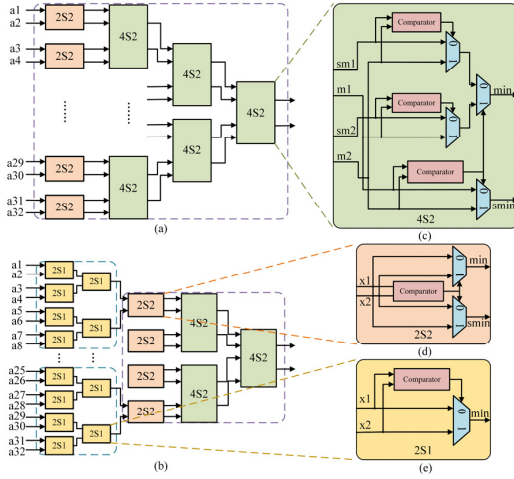


Fig. 1. Search Scheme. (a) TS scheme, (b) grouped scheme; (c) four-input two-output sorter; (d) two-input sorter; (e) two-input comparator.

node c_j , excluding v_i . Each variable node v_i then updates the message sent to the check node c_j by accumulating messages from all other connected check nodes:

$$L_{vi \rightarrow c_j}^{(t)} = L_{vi \rightarrow c_j}^{(0)} + \sum_{ca \in N(v_i) \setminus c_j} M_{ca \rightarrow vi}^{(t)} \quad (3)$$

where $N(v_i) \setminus c_j$ denotes all variable nodes connected to check node v_i , excluding c_j . The updated LLR for variable node v_i is given by:

$$L_{vi}^{(t)} = L_{vi}^{(0)} + \sum_{ca \in N(v_i)} M_{ca \rightarrow vi}^{(t)} \quad (4)$$

A hard decision is made according to:

$$\hat{x}_i = \begin{cases} 0, & \text{if } L_{vi}^{(t)} \geq 0 \\ 1, & \text{otherwise} \end{cases} \quad (5)$$

The decoding terminates when all parity checks are satisfied or the maximum iteration count is reached.

B. Grouped Approximate Minimum Search Scheme

A key challenge in hardware implementation of the NMSA lies in check node processing. Identifying the minimum and second minimum values is computationally demanding for high-degree check nodes, such as those in the CCSDS C2 code, where each node handles 32 inputs. Existing approaches can be broadly classified as sorting-based (XS) or tree-structured (TS) architectures, with TS generally preferred due to lower latency and area requirements [12]. Nevertheless, scalability remains an issue as node degree increases.

To address this issue, we propose a grouped approximation method that partitions the M incoming messages into N groups $\{L_1^{(t)}, L_2^{(t)}, \dots, L_N^{(t)}\}$. This approach reduces comparator count and processing delays by limiting the comparison scope per operation. The core idea, illustrated in Fig. 1 (a) and (b), involves partitioning messages into smaller groups and processing them hierarchically. Each group contains M/N messages. Within each group, the local minimum is computed independently, which reduces the scale of each comparison operation. After intra-group processing, the N local minima are further compared through a tree-structured comparator

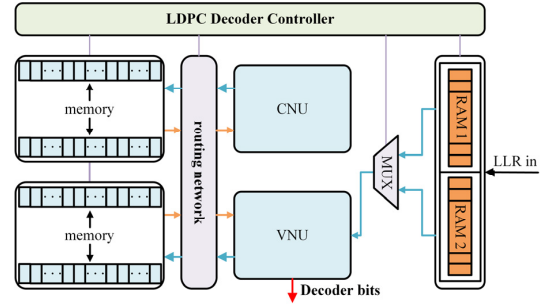


Fig. 2. The architecture of the proposed decoder.

network to obtain a global minimum (min1_global). The approximate second minimum (min2_approx) is then determined by finding the second smallest of the N local minima. This approximation avoids the complexity of finding the global second minimum.

Our design employs three custom-designed blocks: a two-input comparator (2S1), a two-input sorter (2S2), and a four-input, two-output sorter (4S2). The intra-group stage requires $M - N$ 2S1 units to find a minimum, while the inter-group stage uses $N/2$ 2S2 units and $\text{ceil}(N/4)$ 4S2 units to find the min1_global and the min2_approx , resulting in a total of $M + N - 3$ comparators and $M + N - 4$ MUXs. Conventional tree-based sorters typically require $2M - 3$ comparators and $3M - 4$ multiplexers. In contrast, the proposed method reduces the hardware requirements to $M - N$ comparators and $2M - N$ multiplexers. The reduction in comparator count, along with the simplified associated logic, effectively decreases the depth of the combinational logic, thereby shortening the critical path delay.

C. Frame-Parallel Memory Architecture for LDPC Decoding

Memory organization critically impacts LDPC decoder performance on FPGA platforms, especially in satellite systems with limited BRAM resources. The 18-Kb Xilinx BRAMs, configurable as 36×512 or 18×1024 , supports dual-port access for efficient bandwidth utilization. Leveraging this, a frame-parallel memory architecture is proposed, aligned with the CCSDS C2 LDPC submatrix (511×511). Setting the BRAM depth to 512 achieves 99.8% utilization ($511/512$). Multiple frames share BRAM blocks whose combined width matches the native 36-bit port, enabling simultaneous access without extra logic and achieving high throughput with efficient resource use.

As shown in Fig. 2, the memory space is divided into two subgroups, each storing a complete frame in parallel. These subgroups are alternately scheduled by the CNU and VNU, enabling concurrent frame-level to read and write operations. Since the read/write sequence in each iteration is fixed, pre-loading this order into the routing network guarantees conflict-free scheduling and data distribution without pipeline stalls. A scheduling controller dynamically allocates access timing and uses handshake control for lightweight asynchronous switching between CNU and VNU. As the CNU has a latency two cycles longer than the VNU, inserting two wait cycles in the VNU aligns their access timing and ensures conflict-free memory access without additional buffering or

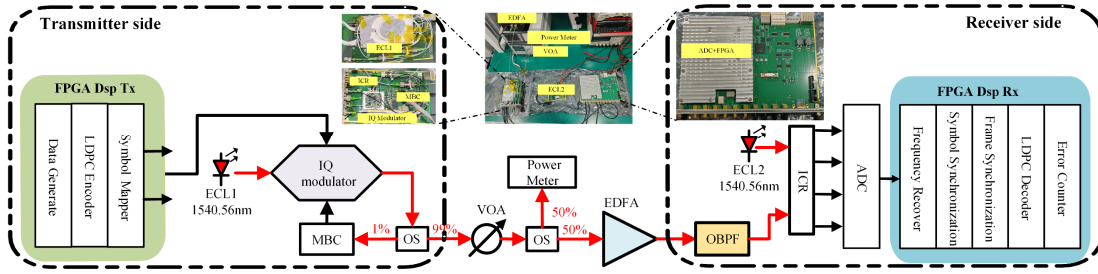


Fig. 3. Experiment framework.

arbitration. The dual-channel memory module concurrently handles LLR and intermediate data, sustaining continuous multi-frame parallelism with conflict-free scheduling and efficient, robust high-throughput operation.

III. EXPERIMENTAL SETUP

The experimental setup, illustrated in Fig. 3, comprises both the optical and digital modules. At the transmitter side, pseudo-random binary sequence (PRBS) data generated by the FPGA is first LDPC-encoded and subsequently mapped using BPSK before being transmitted at symbol rates of 2 Gbps, 1 Gbps, or 500 Mbps. The optical front end comprises an external cavity laser (ECL) operating at a wavelength of 1540.56 nm. The modulated optical signal is launched into the optical link, where a variable optical attenuator (VOA) is employed to precisely control the received power level, and a 50:50 optical splitter (OS) enables simultaneous monitoring of the transmitted optical power for system characterization.

At the receiver side, the incoming optical signal is captured by a heterodyne coherent receiver consisting of a secondary ECL, a pre-amplifier EDFA, and an optical bandpass filter for out-of-band noise suppression. The filtered optical signal is subsequently detected by an integrated coherent receiver (ICR) and digitized by a 12-bit high-speed analog-to-digital converter (ADC) operating at 10GS/s, thereby ensuring sufficient quantization precision for coherent demodulation. Real-time digital signal processing implemented on a Xilinx Virtex-7 (XC7VX690T) FPGA performs frequency offset estimation, carrier phase recovery, and frame synchronization to ensure accurate alignment prior to decoding. The recovered baseband data are then processed by the proposed LDPC decoder, and the resulting decoded sequence is compared with the transmitted PRBS to evaluate the bit-error-rate (BER) performance. Hardware resource utilization is also evaluated and benchmarked against conventional architectures to assess overall decoding efficiency and implementation costs.

IV. EXPERIMENTAL RESULTS

Fixed-point arithmetic is widely used in LDPC decoder implementations to reduce the hardware cost while maintaining performance. MATLAB-based simulations were conducted to evaluate the different fixed-point configurations. The fractional width primarily affects the waterfall region, while the integer width determines the error floor. With a 12-bit ADC resolution, the total word length is limited to 16 bits to ensure adequate margin without loss of precision. With the normalization factor optimized at 0.75, simulations varying the fractional

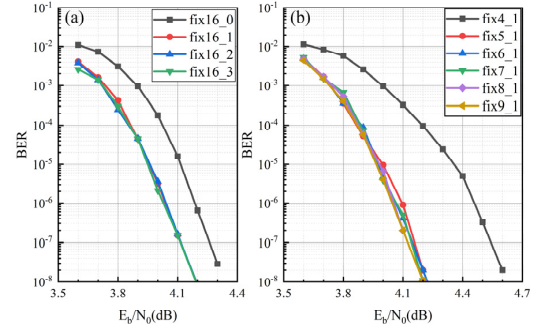


Fig. 4. Decoder BER performance for various fixed-point configurations; (a) effect of fractional width; (b) effect of total word length.

width from 0 to 3 reveal that a single fractional bit yields a 0.2 dB gain over the integer-only scheme. Since further precision offers negligible improvement, a 1-bit fractional width was adopted (Fig. 4(a)).

Next, with the fractional width fixed at 1 bit, the total word length is swept from 4 to 9 bits (fix4_1 to fix9_1). BER results (Fig. 4(b)) show steady improvement up to fix5_1, beyond which gains saturate. Although fix9_1 performs slightly better, its larger data width limits stored frames under fixed RAM width, reducing throughput to 66.7% of fix6_1. For tight resource budgets, fix4_1 remains viable with moderate degradation. Maintaining a performance margin is important to mitigate real-world impairments, such as residual ADC quantization noise or minor channel mismatches. The fix6_1 configuration operates comfortably in the saturated region, whereas fix5_1 lies near the ‘knee’ of the curve, making it more sensitive to input LLR variations. Therefore, fix6_1 is selected as the final configuration, balancing the robustness, hardware efficiency, and decoding throughput.

We further investigated how check-node grouping affects the trade-off between the decoding performance and hardware complexity, evaluating the ROP and resource consumption at various communication rates and group sizes shown as Fig. 5. Decoder performance significantly improves with decreasing data rates. At a fixed rate, a clear trade-off emerges increasing the number of groups improves decoding performance (i.e., lowers the required ROP), but at the cost of diminishing hardware resource savings. Using two groups limits the ROP penalty to approximately 0.1 dB, while achieving comparator and MUX resource savings of up to 49% and 65%. Four groups reduce the ROP penalty to 0.05 dB while maintaining approximately 46% and 60% comparator and MUX resource savings, providing a balanced choice for moderately constrained systems. Eight groups achieve a penalty of <0.02 dB

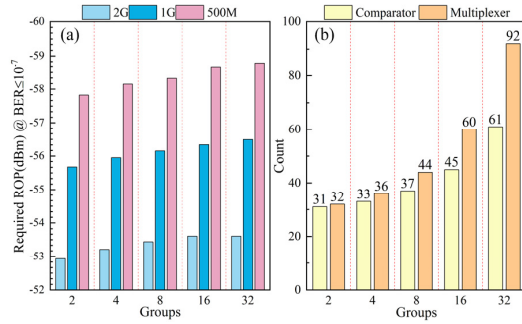


Fig. 5. Experimental results: (a) effect of code rate and node grouping on required ROP at post-LDPC BER $\leq 10^{-7}$; (b) effect of single CNU grouping on comparator and MUX resource utilization.

TABLE I
IMPLEMENTATION RESULTS

work	[11]	[12]	[14]	This paper
Processor	Virtex-5 XC5VLX330	Zyng 7000	Kitnax UltraScale XCKU060	Virtex-7 XC7VX690T
Code length	8176	8176	8176	8176
Code rate	7/8	7/8	7/8	7/8
Algorithm	AMSA	AMSA	AMSA	NMSA
Iterations	10	15	10	10
CLK(MHz)	250	210	350	350
LUTs	46294	38499	48353	35719
FFs	39103	50674	49768	38118
BRAMS	88	144	98	80
Throughput/Gbps	1.2	1.1	2.67	2.87

while maintaining savings of 39% and 52%, making them ideal for high-throughput, resource-constrained designs. At 16 groups, ROP is nearly identical to the ungrouped baseline, but resource savings drop to 26% and 35%, indicating diminishing returns. These results confirm that a moderate grouping strategy can yield significant resource savings at the cost of a small, controllable degradation in performance. At a 1dB link margin, a loss of 0.02dB is negligible, so the 8-group configuration is chosen to achieve the best balance between the resource efficiency and near-benchmark performance.

Following the completion of place-and-route, the proposed decoder operates at a 350 MHz clock frequency. With a 10-iteration schedule, it achieves a peak throughput of 2.87 Gbps, while utilizing a mere 35,719 LUTs, 38,118 FFs, and 80 BRAMs (Table I). In comparison to the designs in [11] and [12], our implementation demonstrates a throughput that is 2.4x and 2.6x higher, respectively. This enhanced performance is primarily attributed to the 350 MHz clock frequency, which represents a significant improvement over the approximately 200 MHz reported in those works, concurrently utilizing fewer logic resources. In contrast to [14], which achieves 2.67 Gbps at approximately 1.2x higher LUT/FF usage, our design attains 2.87 Gbps with superior hardware efficiency. Specifically, it achieves 80.4 Mbps/kLUT (2870 Mbps / 35.7 kLUTs), representing a 25% improvement over the 64.3 Mbps/kLUT reported in [14]. While incurring higher latency (9,447 cycles), this trade-off prioritizes the critical hardware efficiency for bulk transmission. These results demonstrate that the co-design of the grouped approximation and frame-parallel architecture achieves both high throughput and efficiency, thereby

rendering it highly suitable for power- and area-constrained satellite receivers.

V. CONCLUSION

This work presents an efficient LDPC decoder architecture tailored for satellite optical wireless communication systems with stringent resource and performance constraints. By introducing a grouped approximate min-sum algorithm, the design reduces comparator usage by up to 52% and effectively shortens the critical path. Combined with intra-frame pipelining, inter-frame resource reuse, and a compact multi-frame parallel memory strategy, the architecture achieves near-optimal BRAM utilization (511/512 blocks) while overcoming traditional throughput and storage bottlenecks. FPGA implementation results demonstrate reliable operation at 2 Gbps BPSK with a minimum required ROP of -53 dBm and only 0.02 dB performance degradation, thereby validating the proposed algorithm-architecture co-design as a practical and resource-efficient solution for next-generation satellite optical communication systems.

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